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Hypergraph Learning for Fundamental Shape Detection

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Abstract

Geometry was the science of shape and it was used to measure the shape. The shape was the outline of something and it was related to size and pose. There were so many normal and abnormal variability present in the shape of a structure like oval, oblong, round, square, diamond, etc. Shape analysis was an extremely challenging task in medical image processing. There was a need in medical research to abstract the form of an object present in the image. It should describe it with a few geometrical attributes or even with words. And also it should relate it to the form of another object, and group together. All these processes need to be automated. So describing the shape of objects in mathematical representation was the better preference. In this paper, a novel hypergraph based shape recognition framework called MRS-Framework was developed. It was used to identify basic geometric shapes. It will be used in medical analyze to diagnose the disorders.

Keywords: shape analysis, hypergraph, image recognition, medical analysis.

1. Introduction

Euclidean geometry includes the study of points, lines, planes, angles, triangles, congruence, similarity, solid figures, circles, and analytic geometry. Euclidean geometry has practical applications in computer science,

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crystallography, biomedical and various branches of modern mathematics [1]. Few of the shape process and technique will be discussed here. Mathematical based shape definitions were helpful for the programmers to automate the shape analysis model. Shape operator was introduced to find the curves [2,3,4]. Shape detection is also useful in agriculture field. The shape of the flower was acknowledged to detect differences in the aspect ratio of petals [5]. In the medical field, the similarity of shape and classification of shape were to be discovered. An elastic shape analysis approach was done to find the curves. There used arithmetic properties and square root functions to evaluate the curve in the image and applied the result on DNA molecule shape [6]. Locating the Fracture in the human bone was a tedious task for physicians, even with the help of computers. The thickness plays a major role. Shape regression was put into operation to perceive the mesh shape. Gaussian process model help to identify the mesh properties. Overall it was a great tool which was able to turn out meshes with patient-specific bone shape from MRI scans [7]. Medicine was a sensitive field. Which require some automated tool to identify the shape to diagnosis disease like Parkinson. A deep brain simulation tool was developed based on a statistical method for the recognition of the changes [8, 9]. Instead of recognizing the boundary, a distributed shape gradient was introduced. it was an efficient method used to detect brain stoke [10]. An approach called shaping point spread functions was implemented on noise-free shape image and succeeded. This effect will be used for photo-simulation, laser and optical processing [11]. Detecting shape boundaries show the way to understand the shape properties. Study of graph and shape were of great interest in shape property. Shape descriptors were constructed to measure the boundary structure [12]. The Distance Transform Network is one of the best shape analyzers. It used the Euclidean distance to detect the natural shape. it was able to remove noise and adequate to the significant parameter of shape variation and point of reference [13]. Binary shape analysis was done to recognize the boundary along with the measure of aspect ratio. In binary shape analysis, the following challenge was covered. The challenge was to calculate the measurement like mean, standard deviation present in the binary image[14]. Shape analysis plays an important role in bioinformatics. Recognizing the shape of protein helps to identify the similarity between the protein structures. Calculating the loaded diameter for the protein structure was done in the model named similarity of 3D protein models [15, 16]. Biomedical images usually contain complex objects, which will vary in appearance significantly from one image to another. Attempting to measure or detect the presence of a particular structure in such images was a difficult task [17].

Here, the presence of a particular structure and the key challenge of the total count of each shape was addressed. Even though numerous approaches were put into practice and basic information was collected through shape analysis, Still, the count of each shape was not been evaluated. In this paper, A MRS framework was introduced which handles the following two tasks for the real-time images. First, it detects the presence of shape. Second, it counts the total number of each shape in the image. By using MRS framework, it is possible to successfully analyze complex images. Complex images give rise to many primitives, which must be linked together correctly to give a successful interpretation. It will help the researchers and end-users in image processing, medical analysis, forensic analysis for their specific needs.

2. Materials and Methods

Digital images were read in the form of a matrix. There was a need to identify the shape in a medical image [18]. Here the medical image was represented as an incidence matrix, which was the structure of the hypergraph [19]. An NxM size matrix was given as an input to the model. It contains one of the basic geometric shapes, where the task is to find out which shape does the matrix has. The matrix also has any other free form shape, where it was considered as an error.

2.1. Shape Recognizing Mathematical (MRS) Framework

In order to identify each shape, a separate logic was followed. The first task is to identify the line segment. A method was created to identify the line by using a hypergraph neighborhood algorithm. The method reads the NxM input matrix, in the form of a vector. A horizontal line segment was identified when the vector has a sequence of input '1' in the row. A vertical line segment was identified when the vector has a sequence of input '1' in the column. Hypergraph model algorithm works by locating the first presented '1' in the vector. if the next cell or a block has '1'. Then it was the horizontal line. If the nth cell or block was '1' then it was vertical line and if N+1 th row was '1', then it is a diagonal line. To ensure it was a line, follow the sequence by computing the starting position and the ending position of the '1', which was also used to measure the length of the line. If none of the above constraints was satisfied then it was a point in the image, which was the basic shape, representing a binary '1' in the single or individual matrix cell or block location.

$$L=\begin{cases} \sum_{i=1}^{4} I\,[x][y+1],[x+1][y],[x+1][y-1],[x+1][y+1]\notin 0 \\ \qquad \text{otherwise}[([x][y])\notin 0] \end{cases} \tag{1}$$

The equation (1) presents the mathematical representation of the line segment and a point. Where x and y were the coordinate locations of the matrix. A case study was conducted to identify the C. elegans was alive or dead with the identified line segment method and in section 3, the results were discussed. The second task is to identify the square. The mathematical representation of the square and circle was given in equation (2).

$$T=\sum_{i=0}^{3}(x+i)(y+i)\notin 0 : i\ is\ binary\ set \tag{2}$$

The third task is to identify the triangle. Anyone missing point in the square was recognized as a triangle. The mathematical representation of the triangle was given in equation (3).

$$T=if\sum_{i=0}^{3}(x+i)(y+i)\ \notin 0\ and\ one\ of\ \{\ (x+i)(y+i)\}\in 0 \tag{3}$$

A sample 3 x 3 matrix of different shapes were shown in table 1. But the framework works for any size of N x M and N x N matrices. The thickness of the line was measured using the parallel line. Considering the varying shape in the starting and ending position of the line. The matrix was evaluated using the improved combination of the above formulas.

Table 1 Sample matrix for images with different line positions.

1	1	1	1	0	0	1	0	0	0	0	1
0	0	0	1	0	0	0	1	0	0	1	0
0	0	0	1	0	0	0	0	1	1	0	0
Horizontal Line			Vertical Line			Left Diagonal			Right Diagonal		

Table 2 Sample matrix for different images in different positions.

0	0	0	1	1	0	1	1	1	0	0	0
0	1	0	1	1	0	1	1	0	0	1	0
0	0	0	0	0	0	1	0	0	0	0	0
Point			Square			Triangle			Parallel Vertical Line		

3. Experiment and Results

A shape recognition mathematical method was developed. It was constructed and called an MRS framework. An example of a 6x 9 size incidence matrix was given as input for the framework to identify the shape and count of shape. Figure 1 shows the stick man face image and the equation 4 shows the incidence matrix of the corresponding image in Figure 1. The MRS framework inferred that there present 3 horizontal lines and 1 vertical line.

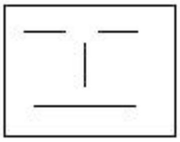


Fig 1 Stick man face image drawn with lines

$$S = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{4}$$

An incidence matrix was considered as an instance. Here, an instance will contain any shape and any number of shapes. The framework was tested for multiple inputs of a different instance of a matrix with varying size. It outperforms in finding the number of basic geometric shapes present in the image. Table 3 shows the various instance generated and the corresponding output of the framework.

Table 3 Sample matrix for images with different line positions.

S.No.	Hypergraph	Number of lines	Number of points	Number of squares	Number of circles	Number of Triangle
1	8x6 &	3	1	0	0	0
2	35X40	0	0	2	0	1
3	60x40	12	0	1	2	3
4	60x60	2	12	3	3	5
5	120x100	3	2	12	7	6
6	200x180	12	0	0	9	0
7	250x240	30	66	5	17	12

Input incidence matrix and the obtained shape information in the image were shown in Figure 2, and it has been inferred that when the matrix size gets an increase, the noise also increases. That is the bigger the size of the medical image leads to accumulation of more noise. The line shape was present in the image with a different angle. Parallel lines were grouped together to find the thickness of the line shape. The ground truth image of C. elegans was collected from Broad Bioimage Benchmark Collection [20]. It was given as input to the MRS framework. The framework identified the shape as follows in Figure 3. A sample square and triangle shape image was randomly chosen from the search engine and tested among the framework. The framework was able to identify the shapes. Point shape is considered as noise in the image. Figure 4 shows the noise image. Separating the noise was an important task in medical images [21]. This framework works as a low level of noise recognition. It was usual that CT, MRI scans and other medical image were with the presence of noise. Even though, the MRS framework outperforms to locate the presence of supervised structure in the image. It needs to be enhanced by considering this noise issue.

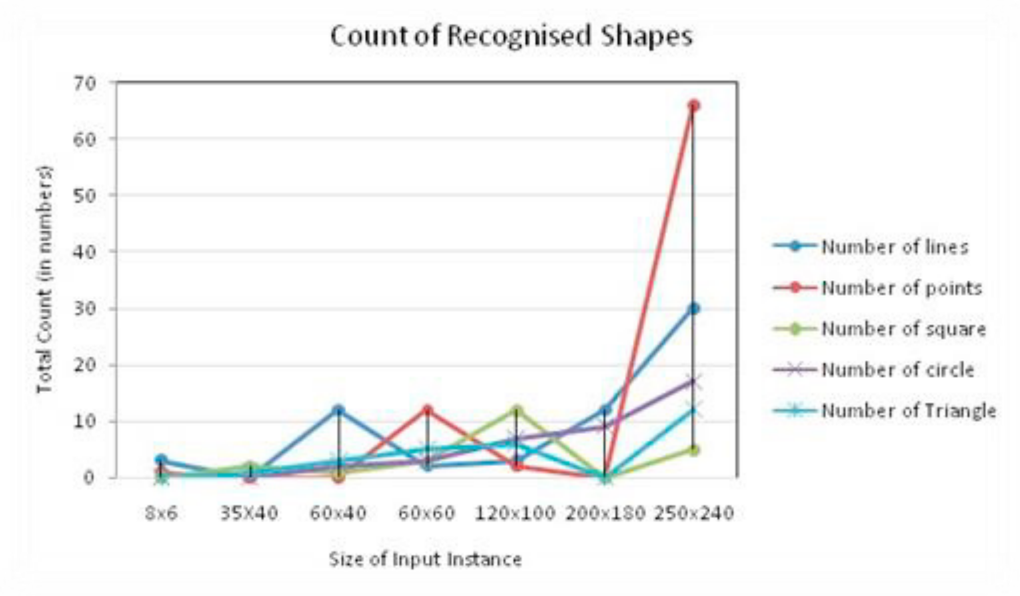
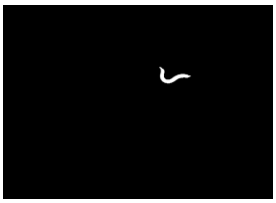


Fig 2 Graph of shape input instance and its count.

(a) Right Diagonal Line.

(b) Vertical Line.

(c) Horizontal Line.



(d) error

(e) Circle with error

. (f) Left Diagonal Line.

Fig 3 Ground truth image of dead C. elegans

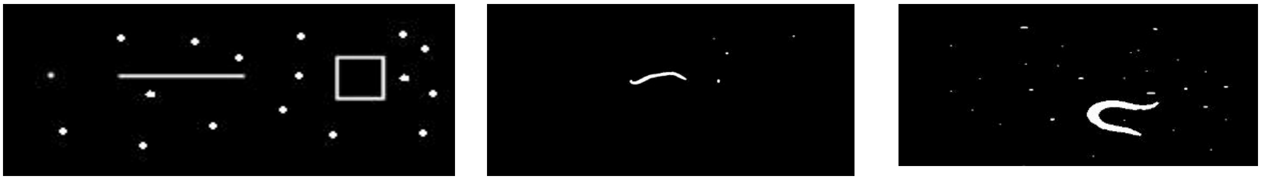


Fig 4 Shapes with noise

3.1. Limitations and Applications

This framework needs to be enhanced with the following medical needs. The brain neurons were represented in the triangle. First, It has to find the triangle shape of brain neuron. Second, it will locate features in images of the human face, the cartilage in MR images of the knee and the shape of the hippocampus. The challenge was that both the issues should be addressed with the hypergraph structure. Third, the framework requires estimating the mean and standard deviation of the identified unsupervised shape. In Brain image analysis, blood clots may disappear over a period of time. By analyzing the patient history of MRI scans, the growth of disease can be identified with merging all the images. It will help the investigators and end-users in image processing, clinical tests, forensic analysis for their specific needs.

4. Conclusion

Basic Shape analysis was done to recognize the shape. The MRS-Framework was used to identify the basic geometric shapes present in the image. It described the identified shape with the geometrical words. In the future, the MRS framework will be extended to categorize the deformable models and will be allowed to use by physicians and researcher for shape analysis in medical image processing.

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